

EXPERIMENT - 8

AIM

To develop a JavaScript-based validation system in the Online Bus Pass Application System that ensures passenger details are correctly entered before submission and then processed using Servlet/JSP.

ALGORITHM

1. Assumptions (Business Logic):

- *Passenger name must not be empty and should contain only alphabets.*
- *Password must be at least 6 characters long.*
- *Email must follow a valid format.*
- *Mobile number must contain exactly 10 digits.*
- *All required fields must be filled before submission.*
- *Form submission should occur only if all validations are successful.*

2. Input:

- *The user enters details such as name, password, email, mobile number, and address in the bus pass registration form.*

3. Steps for Validation Using JavaScript

Step 1: Create the Form (HTML)

- *Design a bus pass registration form with fields like name, password, email, mobile number, and address.*
- *Attach a JavaScript validation function using the onsubmit event.*

Step 2: Capture Form Data

- Use JavaScript methods like `getElementById()` to retrieve user input values from the form fields.

Step 3: Perform Validation Checks

- Check if all fields are filled.
- Validate name (only alphabets).
- Validate password length (minimum 6 characters).
- Validate email format using regular expressions.
- Validate mobile number (exactly 10 digits).

Step 4: Display Error Messages

- Show appropriate alert messages if validation fails.
- Prevent form submission using `return false`.

Step 5: Allow Form Submission

- If all validations are successful, allow the form to be submitted to a Servlet/JSP page for further processing.

Step 6: Server-Side Processing (Servlet/JSP + Database)

- The servlet/JSP receives validated data.
- Perform additional validation (security).
- Connect to the database (e.g., MySQL).
- Store or retrieve user data as required

CODE:

```
<!DOCTYPE html>
<html>
<head>
<title>Online Bus Pass Registration</title>
```

<style>

```
body{  
margin:0;  
font-family:Arial;  
background:linear-gradient(135deg,#1e3c72,#2a5298);  
color:white;  
}
```

```
header{  
background:#0d47a1;  
padding:18px;  
text-align:center;  
font-size:26px;  
font-weight:bold;  
}
```

```
.container{  
width:550px;  
margin:40px auto;  
padding:35px;  
background:rgba(255,255,255,0.15);  
border-radius:18px;  
box-shadow:0 0 25px rgba(0,0,0,.35);  
}
```

```
h2{  
text-align:center;  
margin-bottom:25px;  
}
```

```
input,textarea{  
width:100%;  
padding:12px;  
margin:10px 0;  
border:none;
```

```
border-radius:8px;  
font-size:15px;  
}
```

```
textarea{  
height:80px;  
resize:none;  
}
```

```
button{  
width:100%;  
padding:14px;  
border:none;  
border-radius:30px;  
margin-top:20px;  
background:linear-gradient(45deg,#00e5ff,#00c853);  
color:white;  
font-size:18px;  
cursor:pointer;  
font-weight:bold;  
}
```

```
button:hover{  
transform:scale(1.03);  
}
```

```
</style>
```

```
<script>
```

```
function validateForm(){  
  
var name=document.getElementById("name").value.trim();  
var password=document.getElementById("password").value;  
var email=document.getElementById("email").value.trim();  
var mobile=document.getElementById("mobile").value.trim();  
var address=document.getElementById("address").value.trim();
```

```
var namePattern=/^[A-Za-z ]+$/;
var emailPattern=/^[^ ]+@[^ ]+\.[a-z]{2,3}$/;
var mobilePattern=/^[0-9]{10}$/;

if(name==" " || password==" " || email==" " || mobile==" " || address==" "){
    alert("All fields are required");
    return false;
}

if(!namePattern.test(name)){
    alert("Name must contain alphabets only");
    return false;
}

if(password.length<6){
    alert("Password must be at least 6 characters");
    return false;
}

if(!emailPattern.test(email)){
    alert("Invalid Email Format");
    return false;
}

if(!mobilePattern.test(mobile)){
    alert("Mobile Number must contain 10 digits");
    return false;
}

alert("Validation Successful");
return true;

}

</script>
```

</head>

<body>

<header>

Online Bus Pass Application System

</header>

<div class="container">

<h2>Passenger Registration Form</h2>

<form action="EXP8Servlet"

method="post"

onsubmit="return validateForm() ">

<input type="text"

id="name"

name="name"

placeholder="Enter Passenger Name">

<input type="password"

id="password"

name="password"

placeholder="Create Password">

<input type="text"

id="email"

name="email"

placeholder="Enter Email">

<input type="text"

id="mobile"

name="mobile"

placeholder="Enter Mobile Number">

<textarea

```
id="address"  
name="address"  
placeholder="Enter Address"></textarea>
```

```
<button type="submit">  
Submit Application  
</button>
```

```
</form>
```

```
</div>
```

```
</body>  
</html>
```

```
package WTMA;
```

```
import java.io.*;  
import java.sql.*;  
import jakarta.servlet.*;  
import jakarta.servlet.http.*;  
import jakarta.servlet.annotation.*;
```

```
@WebServlet("/EXP8Servlet")  
public class EXP8Servlet extends HttpServlet {
```

```
protected void doPost(HttpServletRequest request,  
    HttpServletResponse response)  
throws ServletException, IOException {
```

```
    response.setContentType("text/html");  
    PrintWriter pw=response.getWriter();
```

```
    String name=request.getParameter("name");  
    String password=request.getParameter("password");  
    String email=request.getParameter("email");  
    String mobile=request.getParameter("mobile");
```

```
String address=request.getParameter("address");
```

```
try{
```

```
Connection con=DriverManager.getConnection(  
"jdbc:mysql://localhost:3306/lalith?useSSL=false&serverTimezone=UTC",  
"root",  
"root123"  
);
```

```
PreparedStatement ps=  
con.prepareStatement(  
"insert into passengers(name,password,email,mobile,address)  
values(?,?,?,?,?)"  
);
```

```
ps.setString(1,name);  
ps.setString(2,password);  
ps.setString(3,email);  
ps.setString(4,mobile);  
ps.setString(5,address);
```

```
int x=ps.executeUpdate();
```

```
pw.println("<html><head><title>Bus Pass Registration</title>");
```

```
pw.println("<style>");  
pw.println("body{margin:0;font-family:Arial;background:linear-  
gradient(135deg,#0f172a,#1e3a8a);color:white;}");  
pw.println("header{background:#2563eb;padding:18px;text-align:center;font-  
size:24px;font-weight:bold;}");  
pw.println(".card{width:550px;margin:70px auto;padding:35px;border-  
radius:20px;background:rgba(255,255,255,0.12);backdrop-  
filter:blur(10px);box-shadow:0 10px 30px rgba(0,0,0,.4);text-align:center;}");  
pw.println(".ok{font-size:55px;}");  
pw.println(".badge{display:inline-block;background:#22c55e;padding:10px  
18px;border-radius:20px;margin-top:15px;}");
```



```
pw.println("p{font-size:18px;}");  
pw.println("</style>");
```

```
pw.println("</head><body>");
```

```
pw.println("<header>");  
pw.println("Online Bus Pass Application System");  
pw.println("</header>");
```

```
pw.println("<div class='card'>");
```

```
if(x>0){
```

```
pw.println("<div class='ok'> ✓</div>");  
pw.println("<h2>Application Submitted Successfully</h2>");
```

```
pw.println("<p><b>Name:</b> "+name+"</p>");  
pw.println("<p><b>Email:</b> "+email+"</p>");  
pw.println("<p><b>Mobile:</b> "+mobile+"</p>");  
pw.println("<p><b>Address:</b> "+address+"</p>");
```

```
pw.println("<div class='badge'>Stored in Database</div>");
```

```
}
```

```
else{
```

```
pw.println("<h2>Registration Failed</h2>");  
}
```

```
pw.println("</div>");  
pw.println("</body></html>");
```

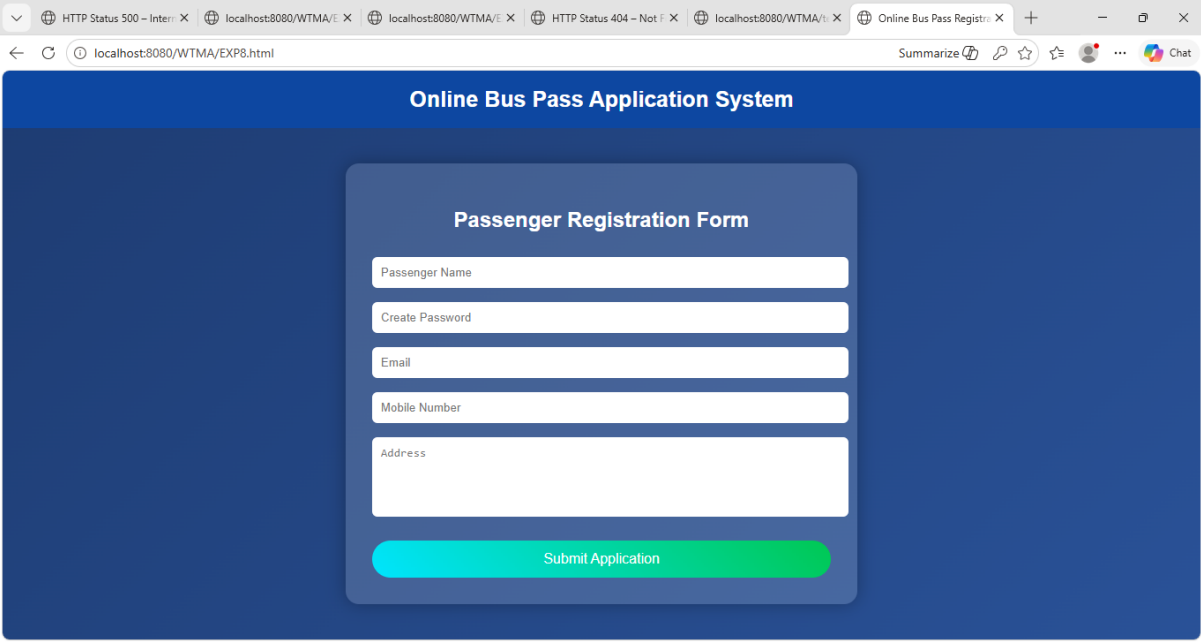
```
con.close();
```

```
}
```

```
catch(Exception e){
```

```
pw.println("<h2>Database Error</h2>");  
pw.println(e);  
  
}  
  
}  
  
}
```

OUTPUT:



The screenshot displays a web browser window with multiple tabs. The active tab is titled 'Online Bus Pass Registr...' and shows a web page with the URL 'localhost:8080/WTMA/EXP8.html'. The page features a dark blue header with the text 'Online Bus Pass Application System'. Below the header is a light blue box containing the 'Passenger Registration Form'. The form includes five input fields: 'Passenger Name', 'Create Password', 'Email', 'Mobile Number', and 'Address'. At the bottom of the form is a green 'Submit Application' button. The browser's address bar shows the URL 'localhost:8080/WTMA/EXP8.html' and the page title 'Online Bus Pass Application System'.

RESULT:

The JavaScript-based form validation was successfully implemented to verify user inputs based on defined business rules. Invalid inputs were prevented from being submitted, ensuring data integrity. Upon successful validation, the form data was processed by Servlet/JSP and stored/retrieved from the database, demonstrating effective client-side and server-side validation.

